

Project 2 - What's My Grade, Really?

Professional Project Report

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AI Tool Used: ChatGPT

Executive Summary

This project converts a teacher-specific grading policy into a reusable Python calculator. It applies weighted categories, dropped scores, a conditional final-exam replacement rule, participation bonus points, and target-grade analysis.

Data Integrity Note

The uploaded workbook explicitly states that its scores are dummy data. The calculations below are accurate for the supplied figures, but the workbook must be updated with real scores before the output can be described as a personal academic result.

Grading Policy

Category	Weight	Rule
Assignments	20%	Drop lowest 2
Quizzes	15%	Drop lowest 1
Midterm 1	15%	Lower midterm may be replaced
Midterm 2	15%	Lower midterm may be replaced
Final Exam	35%	Replaces lower midterm if higher
Participation	Up to 3 pts	Added after weighting; cap 100%

Verified Calculation

Metric	Result
Assignment average after drops	85.875%
Quiz average after drops	85.000%
Adjusted lower midterm	90.000%
Higher midterm	82.000%
Final exam	90.000%
Weighted grade before bonus	87.225%
Participation bonus	2.500 points
Current final grade	89.725%

Manual Verification

The assignment category was recalculated by hand. The percentages were 90%, 75%, 92%, 80%, 60%, 83.333%, 95%, 56%, 86.667%, and 85%. After dropping 56% and 60%, the remaining eight scores totaled 687%. Dividing by eight gives 85.875%, exactly matching the script.

Target Grade Analysis

The final-exam score required to reach a 96% overall grade is 102.55%. This is not achievable under a normal 100% exam cap. Even a perfect final exam would produce a maximum overall grade of 94.725%.

What Worked

The solution correctly applied all weights, drop rules, replacement logic, bonus points, and the overall cap. It also produced transparent intermediate values and identified an impossible target rather than giving misleading advice.

Challenges and Limitations

The principal limitation is the use of dummy data. The calculation is also only as reliable as the supplied policy and does not include unstated extra-credit opportunities.

Conclusion

The project demonstrates a practical client-not-coder workflow: describe the grading rules clearly, have AI encode them, verify one category independently, and use the resulting script for ongoing updates.